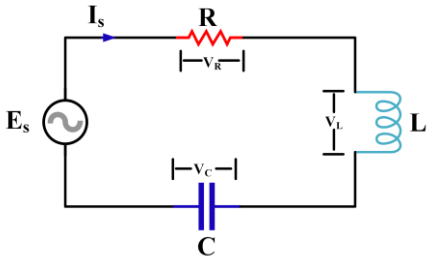
Chapter 52 – RCL Circuits
Calculus Physics II

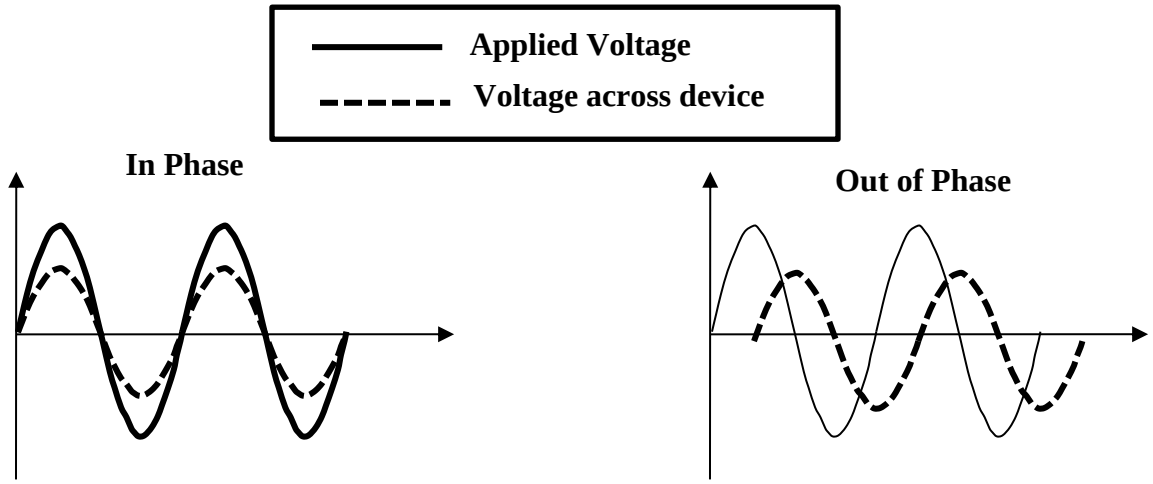
Alternating Circuits – An electrical voltage/current that periodically changes in magnitude and in direction.

- For the case of most electrical devices the current flows as a sinusoidal pattern with time and can be described by the number of cycles the current goes through per second (frequency).
- When there are capacitors and inductors in a circuit there will be a phase shift of the voltage across the devices relative to the applied voltage.



In ac circuits the current through or voltage across the device in the circuit "see" a sinusoidal wave over time. If any of the devices change the phase of the wave, then there can be constructive or destructive interference of the ac waves. This can create changes in voltage/current that the devices "see".

Phase – the position of one wave relative to a second wave. When the waves line up in time then they are in phase, this is constructive interference of waves and will result in a larger wave. When they are not aligned in time they are out of phase, this is destructive interference and will result in smaller wave.

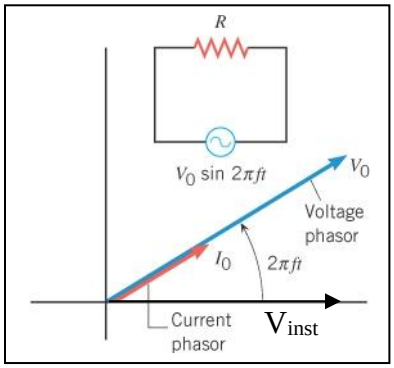


Impedance (Z) – Impedance is the total opposition to the flow of current (charge) in a circuit. The resistance, capacitive reactance and inductive reactance all combine to equal the impedance of the circuit.

Phasor Diagrams – are a graphical representation of the relative magnitude and phase of the current and voltage of an ac component in the real (x-axis) and imaginary (y-axis) plane.

- A Phasor is not a physical quantity with direction in space but is a geometric representation for analyzing quantities that are sinusoidal in time.
- The voltage/current has a real and imaginary component as it rotates in the counterclockwise direction with their base at the origin.

Current and Voltage phase (Phasor Diagram) for Resistor



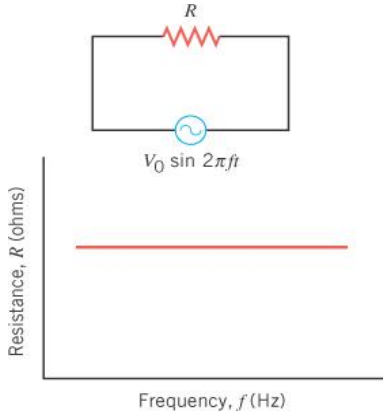
$$V_{inst} = V_o \cos(2\pi ft)$$

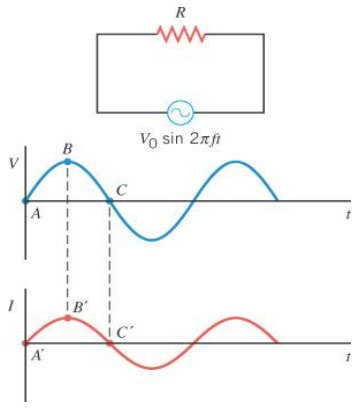
$$V_{rms} = I_{rms} R$$

- The Phasor rotates around the origin at an angular velocity of ω and angular displacement $\Theta = \omega t = 2\pi ft$
- The instantaneous Voltage (V_{inst}) is the projection of the Phasor on the horizontal axis at time t and the real component of the voltage.

Resistors (R) in Alternating Current Circuits - When a resistor is placed in an ac circuit the resistive properties of the resistor do not change due to the changing current or change due to any change in the frequency of the current. Because of the method that resistors use to oppose the flow of the charge, there is a loss of energy in the circuit and this energy is dissipated as heat by the resistors (P = IV).

Voltage and Current in-phase - For a resistor as the voltage changes across the resistor the current changes at the same time and at the same rate. The voltage across the resistor stays in phase with the supplied voltage from the power supply.





Capacitors in Alternating Current Circuits - When a capacitor is placed in an ac circuit it introduces an obstacle to the flow of charge due to charging and discharging of the plates. The magnitude of this reduced flow is proportional to the frequency of the current in the circuit.

- ❖ For very low frequencies the reduction in flow of charge is high (dc current it is zero after capacitor is charged, acts like open circuit).
- ❖ For very high frequencies there is less reduction of flow (at extremely high frequencies the capacitor act as a conducting wire, no capacitance effects).

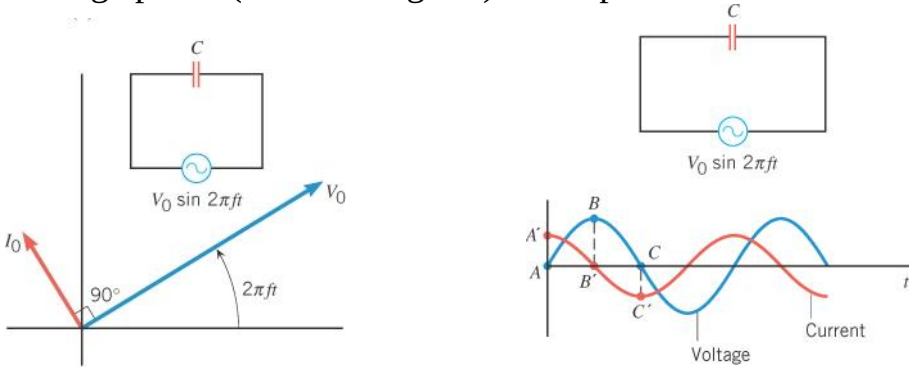
Capacitive Reactance (X_C) A quantity which determines the amount the flow of current is opposed by the capacitor in response to a given rms voltage across the capacitor (units are ohms).

- Capacitive Reactance is a measure of the effectiveness in resisting the flow of an alternating current, by virtue of the potential deference across it owing to the accumulation of charge on its plates.

$$V_{rms} = I_{rms}X_C \qquad X_C = \frac{1}{(2\pi fC)} \quad \text{units of Ohms } (\Omega)$$

- Average Power used by a capacitor in an ac circuit is zero, energy is put into the capacitor as charge (or electric field) and that same energy is released when the capacitor is discharged.

Current and Voltage phase (Phasor Diagram) for Capacitor



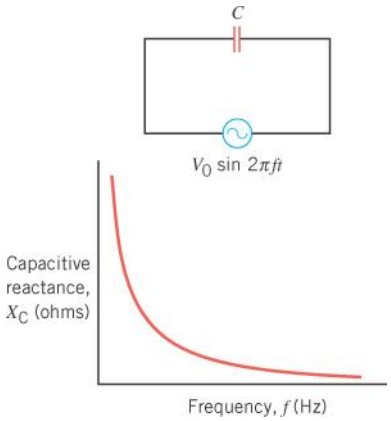
The current leads the voltage for a capacitor driven by an ac power supply

"ICE" – (I) current leads (E) EMF/Voltage in (C) capacitive circuit

For a capacitor in a ac circuit, as the frequency is increased the impedance (resistance to flow of charge) goes down.

- At low frequencies the impedance is very high, very little current flows
- At high frequencies the impedance goes down to almost zero, it is almost like there is no device in the circuit at all.

A capacitor is wired in series with a ac power supply, as the frequency is increased the Capacitive reactance decreases therefore more current flows through the circuit at higher frequencies



Example: Only a capacitor (C = 2 μF) and ac power supply are wired in series.

A) Determine the current through the circuit if the ac supply is running at 12 V and 120 Hz.

$$X_C = \frac{1}{(2\pi fC)} = 663\ \Omega \quad and\ \ I = \frac{V}{X_C} = 0.018\ A$$

B) Determine the current through the circuit if the ac supply is running at 12 V and 1200 Hz.

$$X_C = \frac{1}{(2\pi fC)} = 66.3\ \Omega \quad and\ \ I = \frac{V}{X_C} = 0.18\ A$$

C) Determine the current through the circuit if the ac supply is running at 12 V, 1200 Hz and the capacitor was replaced by a larger 20 μF capacitor.

$$X_C = \frac{1}{(2\pi fC)} = 6.63\ \Omega \quad and\ \ I = \frac{V}{X_C} = 1.8\ A$$

Inductors in Alternating Current Circuits - When an inductor is placed in an ac circuit it introduces an obstacle to the flow of current (charge) due to the induced voltage that opposes the increasing or decreasing current. This reduction in current flow is dependent on the frequency of the current in the circuit.

- For very low frequencies the reduction in the flow of charge is small due to the fact that the change in current is slower so there is less induced voltage to counter the external change. (dc current acts like a conducing wire no induced emf)

- For high frequencies the reduction in flow is very high due to the fast-changing current inducing large induced voltages. (for very high frequencies the induced voltage will equal the applied voltage and no current will flow)

<u>Inductive Reactance</u> (X_L) A quantity which determines the amount of opposition to the flow of current through an inductor in response to a given rms voltage across the inductor (units are ohms).

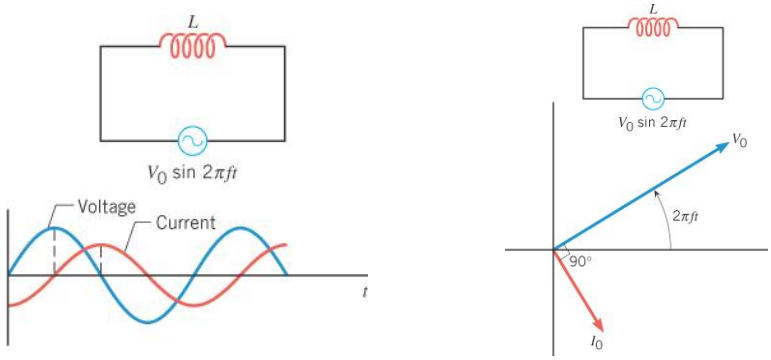
- Inductive Reactance is a measure of its effectiveness in resisting the flow of an alternating current by virtue of the self-induced induced emf that the changing current induces.

$$V_{rms} = I_{rms}X_L$$

$$X_L = 2\pi fL \qquad \text{Units: Ohms } (\Omega)$$

  - The average Power used by a inductor in a ac circuit is zero.
  - The Inductance is proportional to the rate of change of the current, therefore the voltage across the inductor is proportional to the time rate of change of the current.

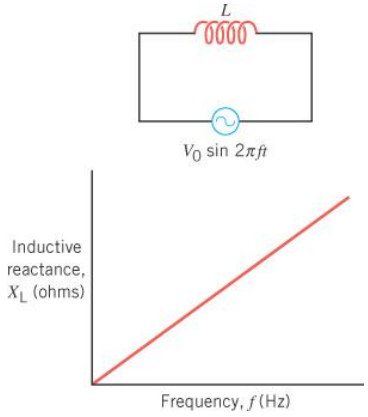
Current and Voltage phase (Phasor Diagram) for Inductor.



"ELI" - (E) EMF/Voltage leads (I) current in (L) inductive circuits

For an inductor in an ac circuit

- The lower the frequency the lower the impedance, there is very little resistance to current flow for low frequencies.
- As the frequency is increased so is the change in current that the inductor sees, this increases the impedance of the inductor in the circuit.



Example: Only an Inductor (L = 2 mH) and ac power supply are wired in series.

A) Determine the current through the circuit if the ac supply is running at 12 V and 120 Hz.

$$X_L = 2\pi fL = 1.5\,\Omega \quad and \;\; I = \frac{V}{X_L} = 8.0\,A$$

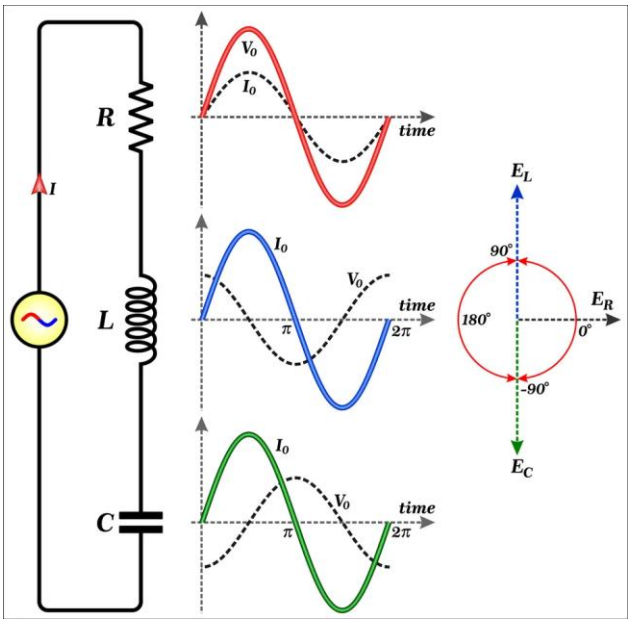
B) Determine the current through the circuit if the ac supply is running at 12 V and 1200 Hz.

$$X_L = 2\pi fL = 15.1\,\Omega \quad and \;\; I = \frac{V}{X_L} = 0.8\,A$$

C) Determine the current through the circuit if the ac supply is running at 12 V, 1200 Hz and the inductor was replaced by a larger 20 mH inductor.

$$X_L = 2\pi fL = 150.8\,\Omega \quad and \;\; I = \frac{V}{X_L} = 0.08\,A$$

Summary



R(f) = R - resistance does not have a frequency dependence.

$X_C = \frac{1}{(2\pi fC)}$ - Capacitive reactance is inversely proportional to the frequency.

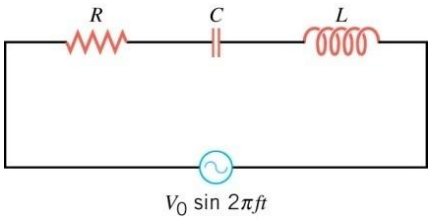
$X_L = 2\pi fL$ - Inductance reactance is inversely proportional to the frequency.

Note from the diagram that when the resistors voltage is along the real axis then the inductor's voltage is along the positive imaginary axis (+90° from resistor – leads) and the capacitor's voltage is along the negative imaginary axis (-90° from resistor - lags).

Series ac Circuits containing Resistance, Capacitance and Inductance (RCL Circuits)

Impedance (Z) – Impedance is the total opposition to the flow of current (charge) in a circuit. The resistance (R), capacitive reactance (X_C) and inductive reactance (X_L) all combine to equal the impedance of the circuit.

$$Z = \sqrt{(R)^2 + (X_L - X_C)^2}$$



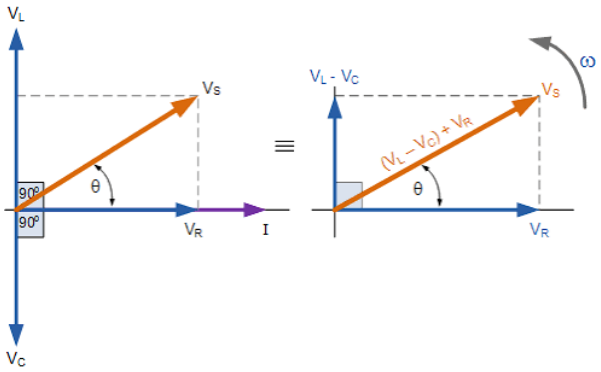
For Series RCL circuits

$V_{rms} = I_{rms}Z$ $\quad Z = [R^2 + (X_L - X_C)^2]^{1/2}$ $\quad \tan\varphi = (X_L - X_C)/R$

The phase angle (φ) in a series RLC circuit represents the time difference, or phase shift, between the total voltage and the total current in the circuit. It indicates whether the circuit's behavior is more inductive (positive angle, current lags), more capacitive (negative angle, current leads), or purely resistive (zero angle) at a given frequency.

The source voltage V_s is equal to the sum of the voltage drops across each component.

$$V_s = V_R + (V_L - V_C)$$



The **complex impedance (Z)** is typically expressed as:

$$Z = R + (X)i$$

Real Part (R): Resistance from resistors.

Imaginary Part (X_i): The net reactance, calculated as the difference between inductive and capacitive reactance.

The imaginary unit $i = \sqrt{-1}$

X is the reactance $X = X_L - X_C$ due to the capacitor and inductor.

The magnitude of the complex form is given as

$$Z = \sqrt{(real)^2 + (imaginary)^2}$$

Therefore

$$Z = \sqrt{(R)^2 + (X_L - X_C)^2}$$

Resonance in Electric Circuits - Resonance occurs when the driving force is equal to the natural frequency of the system. At the resonance frequency the maximum energy is transferred from the source to the circuit. The driving force in the case of the Electrical system is the oscillating electric field.

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

At resonance $X_L = X_C$ and $X_L - X_C = 0$ so

$Z = R$ at resonance

- When resonance occurs the driving force can transmit large amounts of energy to the system.
- For series circuit Impedance is minimum when $X_L = X_C$ under this condition $Z = R$, therefore the resonance of a circuit is that frequency at which $X_L = X_C$ and the phase angle φ is zero

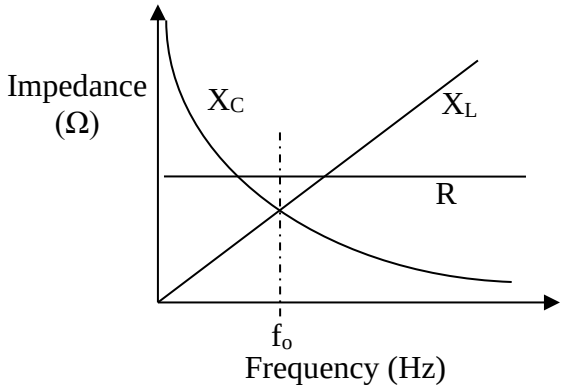
Resonance frequency or RCL circuit (f_{res}) is a condition which occurs when the frequency of the oscillating voltage supplied by a generator matches the natural frequency of the component in the circuit ($X_L = X_C$). Resonance occurs when the driving force is equal to the natural frequency of the system.

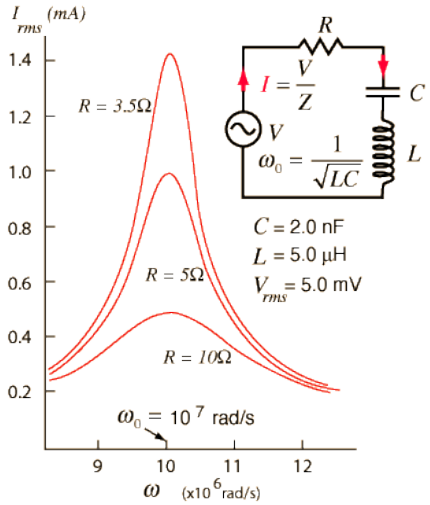
At resonance the impedance (Z) of the series RCL circuit has a minimum value equal to the resistance R and the rms current has a maximum value.

$$f_{res} = \frac{1}{2\pi\sqrt{LC}}$$

- For resonance in series circuits the impedance is minimum therefore the current is maximum. This type of circuit can be used to select a particular frequency.

Impedance and Resonance curves for series RCL Circuit:





The first graph,

- At the resonance frequency the impedance is only the resistance of the resistor because the capacitive reactance is equal to the inductive reactance.
- Resistance has no frequency dependance therefore is constant across the frequencies.
- At lower frequencies the circuit's impedance is more capacitive reactance.
- At higher frequencies the circuit's impedance is more inductive reactance.

The second graph,

- At the resonance frequency the current in the circuit is at maximum.
- The resistance of the resistor only changes the amount of current in the circuit, it does not affect the resonance frequency.
- The values of the capacitor and inductor set the resonance frequency.

Example: A RCL series circuit has a 2 μF capacitor, a 4 mH inductor and a 50 Ω resistor in series with a 100 V rms ac power supply.

A) Determine the current in the circuit if the ac supply has a frequency of 1,000 Hz.

R = 50 Ω $\quad X_C = \frac{1}{(2\pi fC)} = 79.6\ \Omega \qquad X_L = 2\pi fL = 25.1\ \Omega$

$$Z = \sqrt{R^2 + (X_L - X_C)} = 74\ \Omega$$

$$I_{rms} = \frac{V_{rms}}{Z} = 1.35\ A$$

B) Determine the voltage drop across the resistor.

$$V_{rms} = I_{rms}R = 67.5\ V$$

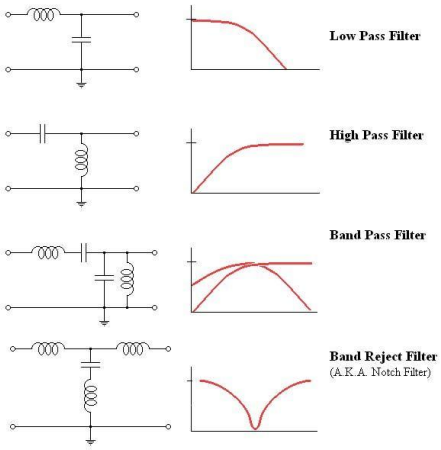
C) Determine the resonance frequency of the circuit.

$$f_{res} = \frac{1}{2\pi\sqrt{LC}} = 1{,}780\ Hz$$

D) Determine the current in the circuit at the resonance frequency.
At the resonance frequency R = Z

$$I_{rms} = \frac{V_{rms}}{R} = 2\ A$$

Electronic High and Low Frequency Filters in ac circuits – when an a.c. circuit is driven by an a.c. power source then



At high frequencies the inductor has the highest impedance

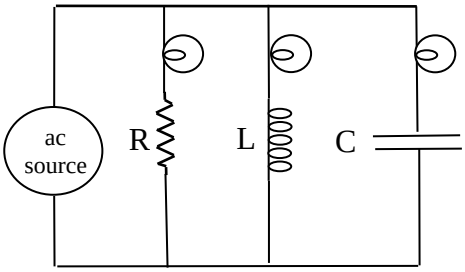
- Inductors are high frequency filters – they reduce the transmission of the higher frequencies while transmitting the lower frequencies.

At lower frequencies the capacitor has the highest impedance

- Capacitors are low frequency filters – they reduce the transmission of the lower frequencies while transmitting the higher frequencies.

The resistor's resistance is unchanged by the frequency of the ac circuit.

Example: Low and High filters

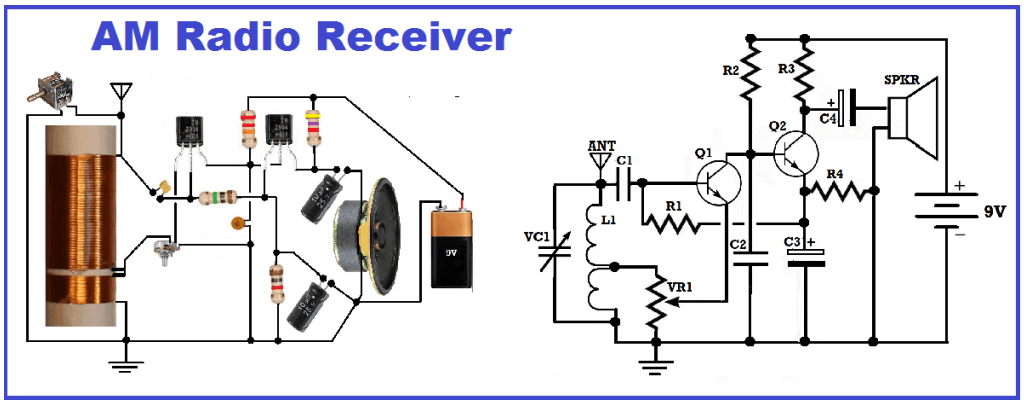


Which light will be the brightest?
a) for high frequency source
b) for low frequency source

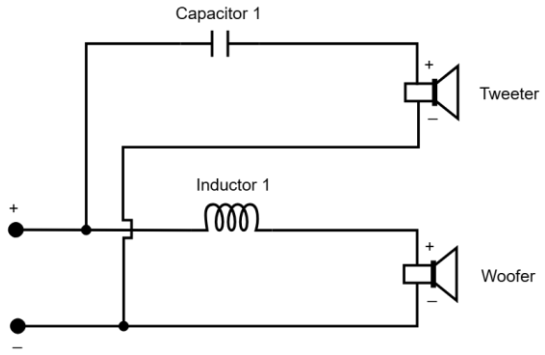
Answer: high f – C brightest
low f – L brightest
bulb R does not change

Application:

Radio tuners use RLC circuits (resistor-inductor-capacitor) to act as a bandpass filter, allowing a specific radio frequency to pass while rejecting others. By adjusting a variable capacitor or inductor with the tuning knob, the circuit's resonant frequency matches the desired station. At this resonance, the impedance is minimal, maximizing current for that signal.

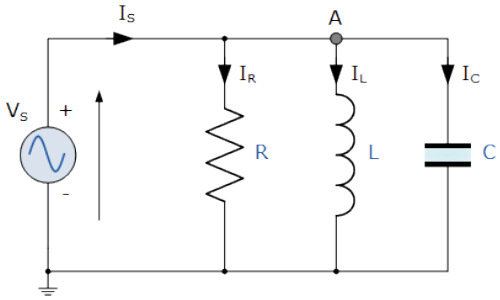


Speaker crossovers utilize passive RLC circuits (Resistor-Inductor-Capacitor) to direct specific frequency ranges to appropriate drivers (tweeters, woofers), acting as frequency-dependent filters. Inductors pass low frequencies (low-pass), capacitors pass high frequencies (high-pass), and resistors attenuate signal levels. These networks prevent distortion by managing impedance peaks, often using parallel RLC circuits to flatten driver impedance resonance.



Parallel RCL Circuits - A parallel RLC circuit consists of a resistor (R), inductor (L), and capacitor (C) all connected in parallel across a common voltage source. Unlike a series circuit where current is constant, in a parallel circuit, the voltage is the same across all components, while the total current is the vector sum of the individual branch currents

$$\frac{1}{Z_p} = \sqrt{\left(\frac{1}{R}\right)^2 + \left(\frac{1}{X_L} - \frac{1}{X_C}\right)^2}$$ For parallel RCL circuit



At resonance $X_L = X_C$ and $X_L - X_C = 0$ so $Z = R$

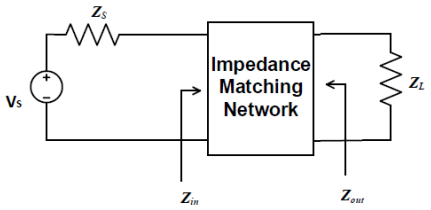
$$f_{res} = \frac{1}{2\pi\sqrt{LC}}$$ same as series RCL

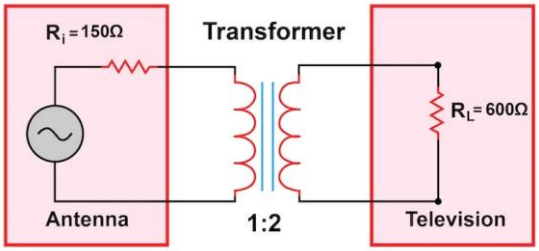
- For resonance in series circuits the impedance is minimum therefore the current is maximum. This type of circuit can be used to select a particular frequency.

Comparison: Series vs. Parallel RLC

Feature	Series RLC	Parallel RLC
Constant Parameter	Current	Voltage
Impedance at Resonance	Minimum	Maximum
Current at Resonance	Maximum	Minimum
Resonant Frequency	Same	

<u>Impedance Matching</u> - In physics, impedance matching is the process of adjusting the electrical impedance of a circuit to maximize the power transfer between a source and a load, or to minimize signal reflections, which is crucial in applications like audio systems and radio frequency (RF) design.





- For a dc circuit, matching is achieved when the load resistance equals the source resistance.

For ac circuits, the load impedance must be the complex conjugate of the source impedance. Impedance matching can be accomplished using matching networks composed of transformers, inductors, capacitors, and resistors.

Maximum transfer of power happens when the impedance of the two circuits are equal. For ac circuits this condition can be achieved by the use of a transformer and satisfying the condition. Where N_1 and N_2 are the winding on the primary and secondary and Z_1 and Z_2 are the impedances of the devices being connected.

$$\frac{N_1}{N_2} = \sqrt{\frac{Z_1}{Z_2}}$$

Example: A power amplifier that has an impedance of 8,000 Ω and a speaker that has an impedance of 8 Ω, hooking them up directly there would be distortion and inefficient power transfer. By using a transformer with a 32:1 winding ratio, the impendence the amplifier "sees" is the same as its impedance of the speaker and there is maximum energy transfer between them.

$$\frac{N_1}{N_2} = \sqrt{\frac{8{,}000\Omega}{8\Omega}} = 31.6$$

Applications:

- Audio Systems: To ensure efficient transfer of power from amplifier to speakers, leading to better sound quality and minimizing distortion.
- Antennas: To efficiently transfer power between a transmitter/receiver and an antenna, maximizing signal strength and range.
- Digital and RF Design: To prevent signal degradation caused by reflections in transmission lines and high-speed circuits.